



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2023 - 2024)

JEE(Advanced)
PRE-SCORE
15-10-2023

JEE(Main + Advanced) : ENTHUSIAST COURSE (PHASE : TOAS, TNAS, TAAS & TTAS)

Time : 3 Hours

PAPER-1 (OPTIONAL)

Maximum Marks : 180

IMPORTANT NOTE : Students having 8 digits **Form No.** must fill two zero before their Form No. in OMR. For example, if your **Form No.** is 12345678, then you have to fill **0012345678**.

READ THE INSTRUCTIONS CAREFULLY

GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **24** pages and that all the **18** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **$g = 10 \text{ m/s}^2$** unless otherwise stated.

QUESTION PAPER FORMAT :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

Please see the last page of this booklet for rest of the instructions

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

SOME USEFUL CONSTANTS

Atomic No. : H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16, Cl = 17, Br = 35, Xe = 54, Ce = 58

Atomic masses : H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24, Al = 27, P = 31, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Br = 80, I = 127, Xe = 131, Ba = 137, Ce = 140

- | | |
|------------------------------------|--|
| • Boltzmann constant | $k = 1.38 \times 10^{-23} \text{ JK}^{-1}$ |
| • Coulomb's law constant | $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$ |
| • Universal gravitational constant | $G = 6.67259 \times 10^{-11} \text{ N-m}^2 \text{ kg}^{-2}$ |
| • Speed of light in vacuum | $c = 3 \times 10^8 \text{ ms}^{-1}$ |
| • Stefan-Boltzmann constant | $\sigma = 5.67 \times 10^{-8} \text{ Wm}^{-2}\text{-K}^{-4}$ |
| • Wien's displacement law constant | $b = 2.89 \times 10^{-3} \text{ m-K}$ |
| • Permeability of vacuum | $\mu_0 = 4\pi \times 10^{-7} \text{ NA}^{-2}$ |
| • Permittivity of vacuum | $\epsilon_0 = \frac{1}{\mu_0 c^2}$ |
| • Planck constant | $h = 6.63 \times 10^{-34} \text{ J-s}$ |

Space for Rough Work

PART-1 : PHYSICS

SECTION-I (i) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

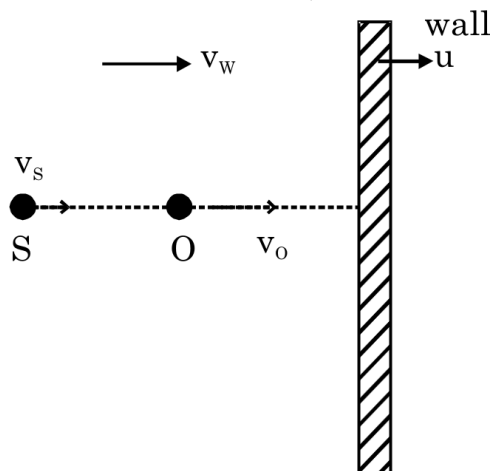
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

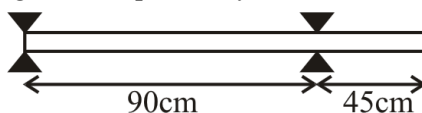
Negative Marks : -2 In all other cases.

- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.
-

1. Consider the situation shown in figure source of sound 'S' and listener 'O' both are moving towards a wall with velocity v_S and v_O respectively. The wall is moving away from them with velocity u . The wind is blowing with a velocity v_W towards the wall. The frequency of sound emitted by source when it is stationary is f_0 . If the velocity of sound relative to air is v , then

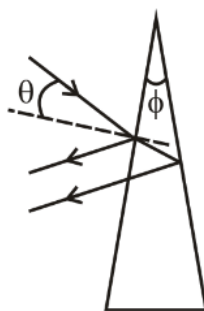


- (A) frequency of sound received by listener after reflection from wall is $\left(\frac{v - v_W + v_O}{v + v_W - v_S} \right) \left(\frac{v + v_W - u}{v - v_W + u} \right) f_0$
- (B) Wavelength of sound received by wall is $\frac{(v - v_S)}{f_0}$
- (C) Wavelength of sound received by listener after reflection from wall is $\frac{(v - v_W + u)(v + v_W - v_S)}{(v + v_W - u)f_0}$.
- (D) Frequency of sound received by listener after reflection from wall $\frac{(v + v_W + u)(v - v_W + v_S)}{(v + v_W - u)} f_0$
2. A metal rod of length 135 cm is clamped at two points as shown in figure. One clamp is at an end and other is at 45 cm from other end. The Young's modulus of elasticity and density of the rod are $Y = 1.6 \times 10^{11} \text{ N/m}^2$ and $\rho = 2.5 \text{ gm/cc}$ respectively.



- (A) the rod will resonate if it is given longitudinal vibrations of frequency 48.84 kHz.
- (B) the rod will resonate if it is given longitudinal vibrations of frequency 26.64 kHz.
- (C) during resonance the number of complete loops between clamps may be 6.
- (D) the maximum wavelength corresponding to vibrations of the rod is 90 cm.

3. Twins Anna and Tom measure each other's heights each birthday. When they were 12 Anna's height was measured as 151.5 cm and Tom's was measured as 151 cm. On their 16th birthday Anna's height is measured as 167.5 cm and Tom's is measured as 168.5 cm. The uncertainty in the height measurements is 5 mm. Anna and Tom have always been very competitive about their heights, so Tom crows with glee after being measured, saying "Finally, I'm taller than you! We were always about the same height, but now I'm taller than you!" Anna responds, "You're crazy! I was clearly taller than you when we were 12." Which of the following statements is most correct?
- (A) Tom is probably right and Anna is wrong, they were about the same height at 12, with a difference of $0.5 \text{ cm} \pm 1.0 \text{ cm}$. Now the measurements show the difference to be $1.0 \text{ cm} \pm 1.0 \text{ cm}$, more accurate measurements would resolve the dispute over who is taller now.
- (B) Tom is probably right and Anna is wrong, they were about the same height at 12, with a difference of $0.5 \text{ cm} \pm 1.0 \text{ cm}$. Now the measurements show the difference to be $1.0 \text{ cm} \pm 1.0 \text{ cm}$, however, taking more measurements, regardless of how accurate, couldn't help resolve the dispute over who is taller now.
- (C) Anna is right and Tom is wrong, Anna was taller at 12, with a difference of $0.5 \text{ cm} \pm 0.5 \text{ cm}$. Now Tom is taller by $1.0 \text{ cm} \pm 0.5 \text{ cm}$.
- (D) Anna is right and Tom is wrong, she was 0.5 cm taller than Tom when they were 12. Tom is now 1.0 cm taller than Anna.
4. A soap film, with constant refractive index $\mu = 1.5$, is contained within a vertical rectangular wire frame and has a wedge shaped profile, small wedge angle ϕ , figure. The film drains under gravity, being thicker at the bottom than at the top, where the thickness can be considered to be zero. The film is viewed, by reflection, with red and violet light at near normal incidence; the angle of incidence being exceedingly small. Interference results from light reflected from the outer surface and that emerging after reflection at the inner surface of the film. The first violet, wavelength 420 nm, constructive interference band is observed at a distance $x_v = 3.0 \text{ cm}$ from the top edge of the film. The wavelength of the red light is 700 nm.

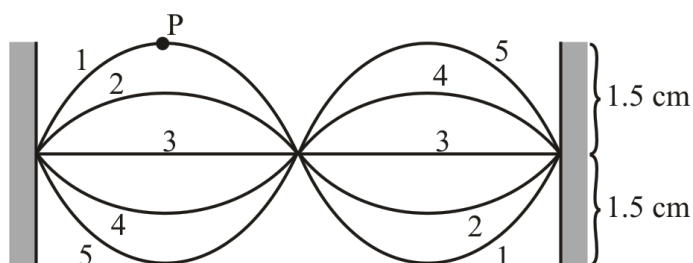


- (A) The thickness of the film at the lowest order constructive violet interference is 70 nm.
- (B) The film wedge angle ϕ is 2.33×10^{-6} radian.
- (C) For the red light, the distance of lowest order constructive interference from the top of the film is 2.5 cm.
- (D) The soap film drains resulting in a decrease in wedge angle at a rate of 8×10^{-5} rad/min. The initial speed of observed red interference fringes is 5.7 cm/s.

5. For the data given in the table below.

medium	density kg m^{-3}	sound speed ms^{-1}
air	1.2	300
concrete	2400	3600
water	1000	1500

- (A) If sound is incident from concrete to air, then the percentage of energy reflected at the boundary is approx. 80%.
- (B) If sound is incident from air to concrete, then the percentage of energy reflected at the boundary is approx. 70%.
- (C) If sound is incident from concrete to water, then the percentage of energy reflected at the boundary is approx. 50%.
- (D) If sound is incident from concrete to water, then the percentage of energy reflected at the boundary is approx. 40%.
6. A vibrating string 50.0 cm long is under a tension of 1.00 N. The results from five successive stroboscopic pictures are shown in figure. The strobe rate is set at 1000 flashes per minute, and observations reveal that the maximum displacement occurred at flashes 1 and 5 with no other maxima in between.



- (A) The time period of oscillation of string is 0.24 sec.
- (B) The speed of travelling waves on the string is $\frac{25}{24}$ m/s.
- (C) The mass of the string is 0.92 kg.
- (D) The point P at anti-node moves with a speed of 6.25π cm/sec when the string is at position-3.

SECTION-I (ii) : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- This section contains **TWO** paragraphs.
- Based on each paragraph, there are **TWO** questions
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If **ONLY** the correct option is chosen.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks : -1 In all other cases.

Paragraph for Questions 7 and 8

The spherometer is used to measure thickness of thin plates and radius of curvature of spherical surfaces. The principle of measurement in the spherometer is same as that of the screw gauge. The circular scale of the spherometer is rotated by a right-handed screw. When the circular scale rotates in the counter-clockwise direction it moves up. Before making any measurement, the three legs of the circular scale are placed on a plane level surface and the screw is rotated so that its tip touches the surface. In this position, if the zero mark of the main scale does not coincide with the zero mark of the circular scale, the instrument is said to possess a zero error. If the zero mark of the circular scale lies a little ahead (in the direction of rotation) of that of the main scale, the zero error is taken to be positive. If the zero mark of the circular scale lies a little behind that of the main scale, the zero error is negative. In the spherometer shown in fig(a), the total number of divisions on the circular scale of the spherometer is 200 and 5 divisions of the main scale is equal to 10 mm. When the spherometer is given one rotation it travels a distance of one division on the main scale. This is called the pitch of the spherometer. In the level position of the legs and the screw, the zero mark of the circular scale lies 50 divisions (of the circular scale) behind the zero mark of the main scale. When thickness of the plate is measured, 45th division of the circular scale lies in between 2nd and 3rd division of the main scale.

7. The thickness of the plate in figure (a) is

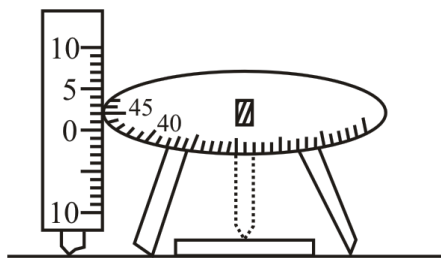


Figure (a)

- (A) 4.45 mm (B) 4.95 mm
(C) 3.95 mm (D) none of these
8. Another spherometer shows the scale, the screw and all the three legs touch the plane surface as shown in figure (b). The least count of the spherometer is given to be 0.01 mm. Then zero error in the instrument is

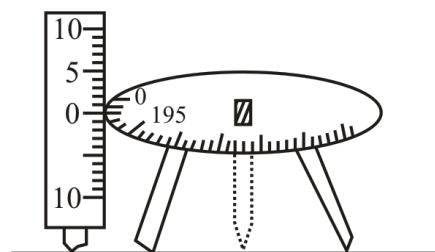


Figure (b)

- (A) 0.98 mm (B) -0.98 mm
(C) -0.02 mm (D) 0.02 mm

Paragraph for Questions 9 and 10

A colloid consists of particles of one type of substance dispersed in another substance. Suspensions of electrically charged microspheres (microscopic spheres, such as polystyrene) in a liquid such as water can form a colloidal crystal when the microspheres arrange themselves in a regular repeating pattern under the influence of the electrostatic force. Colloidal crystals can selectively manipulate different wavelengths of visible light. Just as we can study crystal-line solids by using Bragg reflection of x rays, we can study colloidal crystals through Bragg scattering of visible light from the regular arrangement of charged microspheres. Because the light is traveling through a liquid when it experiences the path differences that lead to constructive interference, it is the wavelength in the liquid that determines the angles at which Bragg reflections are seen. In one experiment, laser light with a wavelength in vacuum of 650 nm is passed through a sample of charged polystyrene spheres in water. A strong interference maximum is then observed when the incident and reflected beams make an angle of 39° with the colloidal crystal planes.

9. Why is visible light, which has much longer wavelengths than x rays do, used for Bragg reflection experiments on colloidal crystals?
- (A) The microspheres are suspended in a liquid, and it is more difficult for x rays to penetrate liquid than it is for visible light.
 - (B) The irregular spacing of the microspheres allows the longer wavelength visible light to produce more destructive interference than can x rays.
 - (C) The microspheres are much larger than atoms in a crystalline solid, and in order to get interference maxima at reasonably large angles, the wavelength must be much longer than the size of the individual scatterers.
 - (D) The microspheres are spaced more widely than atoms in a crystalline solid, and in order to get interference maxima at reasonably large angles, the wavelength must be comparable to the spacing between scattering planes.
10. When the light is passed through the bottom of the sample container, the interference maximum is observed to be at 41° ; when it is passed through the top, the corresponding maximum is at 37° . What is the best explanation for this observation?
- (A) The microspheres are more tightly packed at the bottom, because they tend to settle in the suspension.
 - (B) The microspheres are more tightly packed at the top, because they tend to float to the top of the suspension.
 - (C) The increased pressure at the bottom makes the microspheres smaller there.
 - (D) The maximum at the bottom corresponds to $m = 2$, whereas the maximum at the top corresponds to $m = 1$.

SECTION-II : (Maximum Marks: 24)

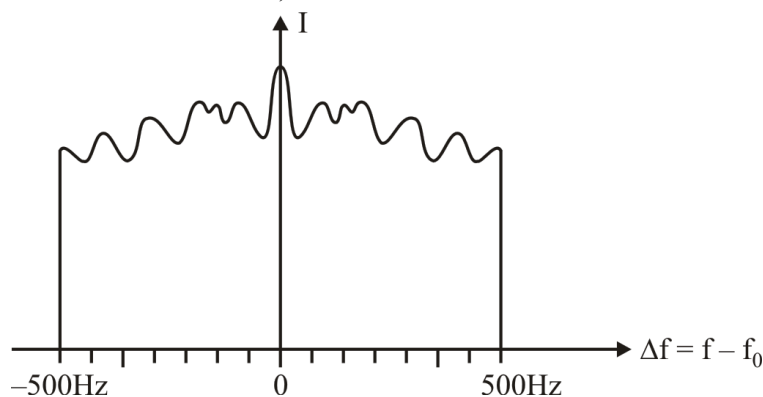
- This section contains **EIGHT (08)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -0.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If ONLY the correct numerical value is entered.

Zero Marks : 0 In all other cases.

-
1. Two tuning forks of frequency 250 Hz and 256 Hz are sounded together. What shall be the time interval (in sec.) between maximum and minimum sound at a point?
 2. A bicycle pump of cross-sectional area 4.0 cm^2 has one end sawn off and a cork is fitted into the end. The piston is pushed slowly inwards and the cork is fired out with a popping sound which has a fundamental frequency of 512 Hz. The initial distance between the cork and the piston is 25 cm, with atmospheric pressure equal to $1.0 \times 10^5 \text{ Pa}$ and the speed of sound in air being 330 ms^{-1} . The force required to eject the cork (in N) is
 3. Monochromatic light of wavelength 600 nm is incident on two vertical slits hence producing two coherent sources. Before the light leaving these slits overlaps and interferes, each beam passes through a tube 5.0 cm long. One of the tubes is now gradually evacuated and it is noted that the fringe pattern shifts 25 fringes. The refractive index of air is more than vacuum by $\alpha \times 10^{-3}$. The value of α is
 4. Two whistles, both of frequency $f = 2.00 \text{ kHz}$, are situated 3.00 m apart and blown simultaneously. An observer travelling along a line parallel to the line joining the whistles, and at a distance approximately 20.0 m opposite the whistles, detects minima in sound intensity at a series of points spaced 1.14 m apart. Calculate the speed (in m/s) of sound in air.

5. Determining the wind speed in a tornado using conventional meteorological instruments is difficult (since the tornado is small in size and moves) and unsafe. It is proposed to measure it from afar using a portable radar, since there is a lot of dust and small objects inside the tornado that reflects ultrasonic waves. Radar emits ultrasonic waves at a frequency $f_0 = 30$ kHz. The spectrum of the signal reflected from the tornado is shown in the figure. Find the maximum wind speed (in m/s) in the tornado. (Take : speed of sound = 320 m/s)



6. The vernier of a vernier scale is divided into 10 divisions which coincide with 9 division of the main scale, each main scale division being 0.5 mm. When the two jaws of the instrument are in contact with each other, the 4th division of the vernier scale coincides with a main scale division and the zero of the vernier lies to the right of the zero of the main scale. When a sphere is inserted between the jaws, the zero of vernier scale lies slightly to the right of the 1.8 cm and the sixth vernier division coincides with a main division. The diameter (in cm) of sphere will be :-
7. The objective of telescope has diameter 12.2 cm. The distance (in km) at which two small green object placed 30cm apart can be barely resolved by telescope, assuming the resolution to be limited by diffraction by objective only ($\lambda = 5.4 \times 10^{-5}$ cm)
8. An electromagnetic standing wave in a certain material has frequency 1.20×10^{10} Hz and speed of propagation 2.10×10^8 m/s. The distance between a nodal plane of $\vec{E}$ and the closest nodal plane of $\vec{B}$ (in cm) is :

PART-2 : CHEMISTRY
SECTION-I (i) : (Maximum Marks: 24)

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Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

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- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

-
1. When $\text{Pb}_{10}\text{O}_3(\text{CO}_3)_6(\text{OH})_2$ react with dil. HNO_3 at room temperature it produces :-
 - (A) NO_2
 - (B) PbO_2
 - (C) $\text{Pb}(\text{NO}_3)_2$
 - (D) CO_2
 2. A solid mixture of silver nitrate and an alkali metal nitrate was dissolve in a small volume of water when an excess of an alkali metal salt (Q) was added to the solution, a silver salt precipitated from solution. After the precipitate was collected, dried and weighted, its mass was equal to the mass of the initial solid mixture.
Which of the following substance could be the alkali metal salt (Q) added?
 - (A) Potassium fluoride
 - (B) Sodium cynide
 - (C) Potassium bromide
 - (D) Rubidium iodide

3. The compound(s) having $[\text{Xe}]4f^1$ configuration on metal ion is/are
(Atomic number of Th = 90, Ce = 58, Lu = 71, Pr = 59) :-
- (A) $\text{Na}_3[\text{Ce}(\text{NO}_3)_6]$
- (B) $\text{Na}_3[\text{LuCl}_6]$
- (C) PrO_2
- (D) $(\text{NH}_4)_4[\text{ThF}_8]$
4. $\text{Metal Salt}_{(\text{aq})} + \text{H}_2\text{S}/\text{acidic solution} \rightarrow [\text{Black ppt.}] \xrightarrow[\text{dil. HNO}_3]{\Delta \text{ with}} \text{Black ppt disappear}$
Metal Salt_(aq.) can be :-
- (A) $\text{Pb}(\text{CH}_3\text{COO})_2$
- (B) NiCl_2
- (C) CuSO_4
- (D) $\text{Hg}_2(\text{NO}_3)_2$
5. Diatomic paramagnetic gas is produce by :-
- (A) Reaction of NH_3 & oxygen using Pt/Rh catalyst
- (B) Thermal decomposition of KClO_3
- (C) Heating ammonium nitrite
- (D) Dissolving gold in aqua regia
6. The electrons in _____ are too strongly bound to get excited by flame.
Select correct to fill the blank space.
- (A) Lithium
- (B) Beryllium
- (C) Barium
- (D) Magnesium

SECTION-I (ii) : (Maximum Marks: 12)

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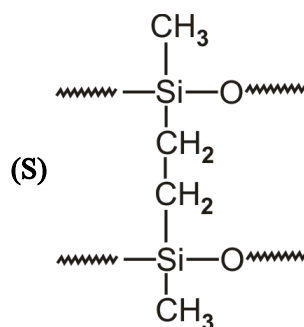
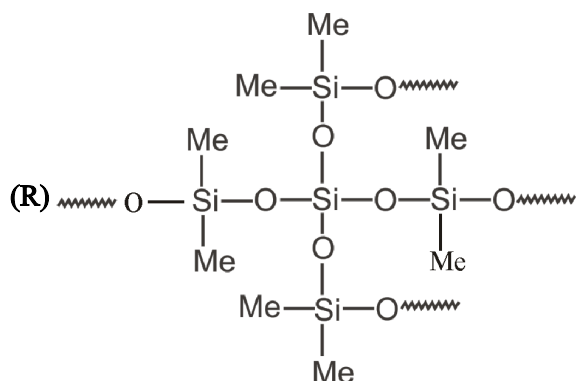
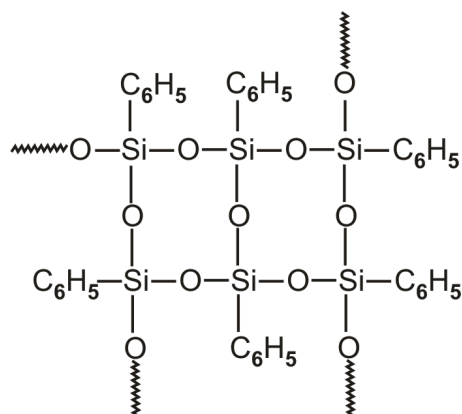
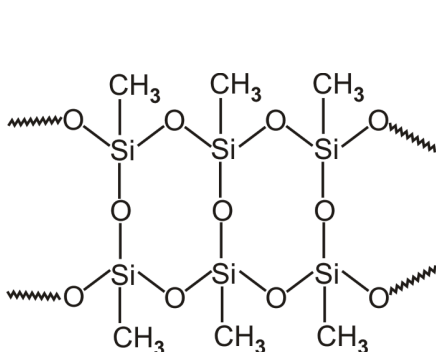
Paragraph for Questions 7 and 8

The substituted Halosilanes undergoes hydrolysis to corresponding substituted Silanols, which on condensation produce silicones. The properties of these silicones depends upon the degree of polymerization and stereo chemistry more -OH groups in the Silanols naturally lead to condensation in one or more direction.

7. Which polymeric substance not required Mono Alkyl/Aryl Silanols during its formation :-

(P) Silicon Resins

(Q) Cross link silicone



Select most appropriate option :-

(A) (P) & (Q)

(B) (R) & (S)

(C) (Q), (R) & (S)

(D) (P) & (R)

8. Which substance has structural resemblance with "Pyrosilicate" ion :-
- (A) Polydimethylsiloxane silicone polymer
- (B) Condensation product of trimethylsilanols
- (C) Condensation product of monoalkyl silanols
- (D) $(\text{Me}_2\text{SiO})_n$ (where $n = 3$)

Paragraph for Questions 9 and 10

Coloured solution (**S₁**) contain two unknown metal ions treated with dil. HCl produces no ppt., Coloured solution (**S₁**) react with H_2S in acidic medium both ions are precipitated, precipitate of both ions are soluble in dil. HNO_3 to gives coloured solution (**S₂**) and turbidity of sulphur.

Solution (**S₂**) produces deep blue solution (**S₃**) & white ppt. (**P₁**) with excess of NH_4OH . After filtration ppt. (**P₁**) dissolve in dil. H_2SO_4 & produce black ppt. (**P₂**) with sodium tetrahydroxidozincate(II).

Solution (**S₃**) first treated with excess dil. H_2SO_4 and then with potassium hexacyanidoferrate(II) produce brown ppt. (**P₃**).

9. Dark Blue solution (**S₃**) is :-
- (A) $[\text{Ni}(\text{NH}_3)_6](\text{NO}_3)_2$
- (B) $\text{Hg}[\text{Co}(\text{SCN})_4]$
- (C) CrO_5
- (D) $[\text{Cu}(\text{NH}_3)_4](\text{NO}_3)_2$
10. Oxidation number of unknown metal ions in precipitate (**P₂**) and (**P₃**) respectively :-
- (A) 0, +2
- (B) +3, 0
- (C) +4, +2
- (D) +5, +3

SECTION-II : (Maximum Marks: 24)

- This section contains **EIGHT (08)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -0.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If ONLY the correct numerical value is entered.

Zero Marks : 0 In all other cases.

- The value of "n" in $P_{(n)}O_{18}^{-6}$ is _____.
The value of "m" in $S_{(m)}O_9$ is _____.
Sum of n & m is your answer.
- Consider given reaction from metal M

$$\begin{array}{c} \text{M} + 2\text{HCl}_{(\text{aq.})} (\text{dil.}) \xrightarrow{\text{Air excluded}} \text{M}_{(\text{aq.})}^{+2} + \text{H}_2 \uparrow \\ \text{(A 3d series metal)} \qquad \qquad \qquad \text{(Blue)} \end{array}$$

$$\text{M}_{(\text{aq.})}^{+2} + 2\text{H}^+ \xrightarrow{\text{slowly}} \text{M}_{(\text{aq.})}^{+3} + \text{H}_2 \uparrow$$

$$\text{M}_{(\text{aq.})}^{+3} \xrightarrow{\text{NaBO}_3 + \text{NaOH}} \text{Y yellow solution (P)}$$

Find sum of the number of d-electrons in $\text{M}_{(\text{aq.})}^{+3}$ & oxidation number of metal in solution (P).
- 100 gram sample of table salt sample which contain 15% of potassium iodate is dissolve in 1 litre of dil. HCl and treated with excess KI.
Calculate amount of H_2O produce in above reaction of iodine containing salts.
(given that atomic mass of K = 39 , I = 127 , O = 16 , Cl = 35.5, H = 1)
- $$\begin{array}{c} \text{(E)}_{(\text{s})} \xrightarrow{\text{KOH}_{(\text{aq.})}} \text{(P)}_{(\text{aq.})} + \text{(Q)}_{(\text{aq.})} \\ \text{(Non metallic element} \\ \text{which has octa – atomic} \\ \text{molecules with puckered ring)} \qquad \qquad \text{salt} \qquad \qquad \text{salt} \end{array}$$

$(\text{P}) + \text{dil. HCl} \rightarrow \text{Gas}-(\text{R}) + (\text{E}) \downarrow$

$(\text{Q}) + \text{dil. HCl} \rightarrow \text{Gas}-(\text{S})$

$(\text{R}) + \text{MnO}_4^-/\text{H}^+ \rightarrow (\text{T}) + \text{Mn contain product}$

$(\text{P}) + \text{NaI}_{3(\text{aq.})} \xrightarrow{\text{H}^+} (\text{U}) + \text{Iodine contain product}$

Find sum of non metal atom (E) – Oxygen linkages (i.e. E –O linkages) in (P), (Q), (R), (S), (T) and (U)

5. • Metal (M_1) of 3d series form unstable MO_4^{2-} in alkaline medium but readily decompose to M_2O_3 & O_2
- Metal (M_2) of 3d series has half filled t_{2g} level in $[M(H_2O)_6]^{+2}$ ion
- Metal (M_3) of 3d series produce stable $M_{(aq)}^{+2}$ ion rather than $M_{(aq)}^{+}$ and $M_{(aq)}^{+}$ is disproportionate to M and $M_{(aq)}^{+2}$.

Identify metal M_1 , M_2 and M_3 and write the sum of atomic number of M_1 , M_2 and M_3 .

6. $HNO_3 \xrightarrow{P_4O_{10}}$ Ionic solid (X) + other product - - -

Ionic solid (X) $\xrightarrow{Na\ metal}$ Ionic solid(Y) + other product - - -

Write the sum of the formula weight of ionic solid (X) & ionic solid (Y).

(Given atomic mass of Na = 23, H = 1, N = 14, O = 16, P = 31)

7. Halogen " X_2 " + Halogen " Y_2 " $\xrightarrow{(Excess)} \xrightarrow{(Z)}$ Interhalogen compound

Z is a orange solid which is dimerises as "Y"-bridged dimer.

Find total number of lone pairs in dimer of "Z".

8. 7.45 gm of KCl (crystals) are heated with excess $MnO(OH)_2$ and $H_2SO_{4(aq)}$. The gas so produced was then passed through a solution of $KI_{(aq)}$. Calculate weight of non polar diatomic solid substance can be produce in final step.

(Given that atomic mass of K = 39, Cl = 35.5, H = 1, O = 16, S = 32, I = 127, Mn = 55)

PART-3 : MATHEMATICS
SECTION-I (i) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

1. If roots of the equation $2x^3 + 4x^2 - 5x + 8 = 0$ are α, β, γ then

(A) $\sqrt{(2-\alpha)(2-\beta)(\gamma-2)(1-\alpha^2)(1-\beta^2)(1-\gamma^2)} = 22.5$

(B) $1+\alpha, 1+\beta, 1+\gamma$ are roots of $2x^3 - 2x^2 - 7x + 15 = 0$

(C) $\sqrt{(2-\alpha)(2-\beta)(2-\gamma)(1-\alpha^2)(1-\beta^2)(1-\gamma^2)} = 22.5$

(D) $1+\alpha, 1+\beta, 1+\gamma$ are roots of $2x^3 - 6x^2 - 7x - 15 = 0$

2. If the value of $\left(1 + \frac{2}{3} + \frac{6}{3^2} + \frac{10}{3^3} + \dots \text{upto } \infty\right)^{\log_{(0.25)}\left(\frac{1}{3} + \frac{1}{3^2} + \frac{1}{3^3} + \dots \text{upto } \infty\right)}$ is ℓ , then ℓ^2 is coprime with

(A) $\left(1 + \tan \frac{\pi}{8}\right)^2$

(B) $\left(\tan \frac{3\pi}{8} - 1\right)^2$

(C) $(\tan 15^\circ - 2)^2 + 1$

(D) $8 \sin 10^\circ \sin 50^\circ \sin 70^\circ$

3. If $f(k) = \sum_{r=1}^k \frac{1}{r}$ and $\sum_{r=1}^{2023} f(r) = af(b) + c$ and $b = 2023$, $a, c \in \mathbb{I}$ then
- (A) $b = c$
- (B) $a - b = 1$
- (C) $b + c = 0$
- (D) $a + c = 0$
4. If α is one of the solutions of the equation $[\log_{10} 4x] = [\log_{10} x]$ such that $\log_{10} \alpha < 0$, then (where $[.]$ denotes greatest integer function)
- (A) Least value of $\frac{1}{\alpha}$ is 4
- (B) Least integral value of $\frac{1}{\alpha}$ is 5
- (C) Greatest value of $\frac{1}{\alpha}$ is 10
- (D) $\frac{1}{\alpha}$ cannot be equal to 11
5. Let $S_n = \frac{4}{3} + \frac{4 \cdot 2^4}{15} + \frac{4 \cdot 3^4}{35} + \frac{4 \cdot 4^4}{63} \dots$ upto n terms. Then
- (A) n^{th} term of sequence is $\frac{4n^4}{(2n+1)(2n+3)}$
- (B) n^{th} term of sequence is $\frac{4n^4}{(2n+1)(2n-1)}$
- (C) $33\{S_{16}\} = 4$ where $\{\cdot\}$ denotes fractional part
- (D) $33\{S_{16}\} = 8$ where $\{\cdot\}$ denotes fractional part
6. Consider $f(x) = ax^2 + bx + c \forall x \in \mathbb{R}$, a quadratic expression such that for $x \in [0, 2]$ range of $f(x)$ is $[0, 1]$ and $-\frac{b}{2a} \notin (0, 2)$
- Then which of following are correct
- (A) The value of C may be 0 or 1
- (B) The sum of roots of $ax^2 + bx + c$ is $P^2 + P + 3$ for $P \in \mathbb{R}$, then $|a|$ can be 2
- (C) The value of $|2a + b|$ is $\frac{1}{2}$
- (D) If sum of roots of $ax^2 + bx + c$ is $P^2 + P + 3$ for $P \in \mathbb{R}$ then $|a| \leq \frac{2}{3}$

SECTION-I (ii) : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- This section contains **TWO** paragraphs.
- Based on each paragraph, there are **TWO** questions
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If **ONLY** the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -1 In all other cases.

Paragraph for Questions 7 and 8

Given that, $A = 2^{\log_3 e^{\log \sqrt{e}^3}}$

$$B = \sqrt{17 + 12\sqrt{2}} - \sqrt{17 - 12\sqrt{2}};$$

$$\text{and } (\sqrt{2} - 1)^{-C} = \frac{4}{A + B}$$

On the basis of above information, answer the following questions :

7. For three series $\frac{A+B}{4}$, λC , $\frac{B-A}{4}$..., select the incorrect option
- (A) They are in G.P. if $\lambda = 1$
 - (B) If they are in A.P. then $\lambda = -\sqrt{2}$
 - (C) If $\lambda = 1$ then sum of infinite terms finitely exists
 - (D) If $\lambda = 1$ then sum of infinite terms is not finite
8. If 2 lies between the roots of quadratic equation $Ax^2 + \sqrt{2}Bx + t = 0$, then $|t|$ can be
- (A) 32
 - (B) 31
 - (C) 33
 - (D) 30

Paragraph for Questions 9 and 10

$f(x)$ is a polynomial of degree n , such that $(\alpha + 1) f(\alpha) - \alpha = 0 \quad \forall \alpha \in \mathbb{N} \cup \{0\}, \alpha \leq n$

9. For $n = 10$; $f(11)$ is

(A) 11

(B) $\frac{10}{12}$

(C) $\frac{10}{11}$

(D) 1

10. For $n = 15$, $f(16)$ is

(A) 15

(B) $\frac{15}{17}$

(C) $\frac{15}{16}$

(D) 1

SECTION-II : (Maximum Marks: 24)

- This section contains **EIGHT (08)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -0.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If ONLY the correct numerical value is entered.

Zero Marks : 0 In all other cases.

- Let A, G^2 and H be the roots of cubic equation $x^3 + px^2 + qx + r = 0$ which are in G.P., where p, q are integers and A, G, H are respectively AM, GM, HM of two positive numbers. If $p, q \in (-100, 100)$ and λ represents the number of all possible ordered triplets (p, q, r) then $\frac{\lambda}{100}$ is
- $\frac{\sum_{r=1}^{20} (r(r+1))^2 (-1)^r}{\left(\sum_{r=1}^{10} (4r)\right)^2}$ is equal to
- Let $T_n = 1 + 2 + 3 + \dots + n$ and $M_n = \frac{T_2}{T_2 - 1} \times \frac{T_3}{T_3 - 1} \times \frac{T_4}{T_4 - 1} \dots \times \frac{T_n}{T_n - 1}$ for $n = 2, 3, 4, \dots$. The value of $52M_{50}$ is
- Let $a_1, a_2, \dots, a_n$, be real numbers such that $\sqrt{a_1} + \sqrt{a_2 - 1} + \sqrt{a_3 - 2} + \dots + \sqrt{a_n - (n-1)} = \frac{1}{2}(a_1 + a_2 + \dots + a_n) - \frac{n(n-3)}{4}$ then find the value of $\sum_{i=1}^{100} a_i$
- If $x = \log_2 3 - 4 \log_{\left(\ell n \frac{5}{4}\right)} 3$, $y = \log_{\sqrt{3}} \left(\ell n \frac{5}{4}\right)$ and $z = \log_2 \left(\log_{(5/4)} e\right)$, then find the absolute value of $(xy + 2z)$
- The minimum value of $x^2 + 2xy + 3y^2 - 6x - 2y$ where $x, y \in \mathbb{R}$ is equal to m. Then sum of all the prime number less than $|m|$ is
- If the equation $x^4 + ax^3 + 2x^2 + bx + 1 = 0$; ($a, b \in \mathbb{R}$) has a real solution, then minimum value of $a^2 + b^2$ is
- Sum of all the real solutions of the equation $x^2 - 3 \left[\sin \left(x - \frac{\pi}{6} \right) \right] = 3$ is m. Then $\frac{[10m]}{2}$ is (where $[.]$ denotes the greatest integer function)

Space for Rough Work

NAME OF THE CANDIDATE

FORM NO.

I have read all the instructions
and shall abide by them.

Signature of the Candidate

I have verified the identity, name and Form
number of the candidate, and that question
paper and ORS codes are the same.

Signature of the Invigilator

Space for Rough Work

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E-24/24

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